

ELHOVO, Bulgaria

WMO: 156420

Lat: 42.183N

Lon: 26.567E

Elev: 456

StdP: 14.46

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.2	18.6	7.9	8.4	16.1	12.5	10.5	21.4	38.6	36.6	37.2	36.4	4.0	0	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.3	93.9	74.3	90.8	72.5	88.1	71.1	77.8	90.0	75.5	86.7	73.2	84.3	6.9	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.9	128.7	86.8	71.7	119.2	83.8	69.5	110.3	80.0	41.6	89.9	39.2	86.6	37.1	84.4	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
27.4	20.8	16.6	7.1	99.5	6.7	4.6	2.2	102.8	-1.7	105.5	-5.5	108.0	-10.4	111.4	
			DB	7.1	99.5	6.7	4.6	2.2	102.8	-1.7	105.5	-5.5	108.0	-10.4	111.4
			WB	6.5	78.2	6.6	5.0	1.7	81.8	-2.2	84.7	-5.9	87.5	-10.7	91.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	34.9	38.5	45.1	53.5	62.2	69.6	74.1	74.3	66.2	56.3	46.4	37.1
	DBStd	15.52	8.72	8.72	7.67	6.39	5.69	4.99	4.35	4.49	5.80	7.22	8.03	9.67
	HDD50	1632	470	328	192	37	2	0	0	0	0	25	164	414
	HDD65	4531	932	741	616	346	119	15	1	1	54	281	560	865
	CDD50	3438	3	7	41	143	381	589	747	753	485	220	55	14
	CDD65	860	0	0	0	1	33	154	283	289	89	10	1	0
	CDH74	9726	0	0	2	34	493	1744	3224	3209	920	99	1	0
	CDH80	3921	0	0	0	4	96	625	1440	1466	275	15	0	0
Wind	WSAvg	4.6	4.3	5.8	6.5	5.4	4.4	3.7	4.2	4.1	4.1	4.3	3.8	4.5
Precipitation	PrecAvg	21.9	1.7	1.6	1.6	1.8	1.9	1.9	1.7	1.1	2.0	2.2	2.1	2.3
	PrecMax	33.5	3.6	5.5	3.3	3.8	4.8	4.7	6.5	3.3	6.0	5.9	5.7	5.3
	PrecMin	14.3	0.1	0.1	0.2	0.3	0.7	0.1	0.0	0.1	0.1	0.1	0.0	0.6
	PrecStd	4.6	1.1	1.3	0.8	0.9	1.2	1.1	1.5	0.9	1.7	1.5	1.5	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.4	63.9	71.1	78.6	86.7	93.5	99.8	98.3	90.7	81.7	71.5	64.6
		MCWB	53.4	53.9	57.2	64.5	68.9	74.6	77.3	76.4	71.6	65.5	60.0	60.5
	2%	DB	54.9	59.7	66.6	73.1	82.7	89.6	94.3	94.3	86.4	76.4	66.5	58.7
		MCWB	50.2	51.3	55.9	60.2	67.6	72.4	74.2	74.8	69.2	64.3	58.8	54.9
	5%	DB	50.9	55.9	62.7	69.6	79.3	86.3	90.7	90.9	83.0	72.5	62.4	55.0
		MCWB	47.3	49.3	53.2	58.1	65.5	70.5	72.3	72.9	67.2	62.0	56.7	51.7
	10%	DB	46.9	51.8	58.6	66.4	76.3	83.5	87.8	87.9	79.4	68.5	58.5	51.3
		MCWB	44.5	46.9	51.0	56.3	63.9	69.0	70.9	71.0	65.4	60.0	54.1	49.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	55.9	59.6	64.7	72.5	79.0	81.6	81.1	74.5	68.5	62.4	60.6
		MCDB	57.9	61.6	67.6	75.6	80.5	90.4	95.1	94.8	85.9	77.2	67.7	63.9
	2%	WB	51.1	52.6	56.4	61.4	69.4	75.6	77.7	77.8	71.0	65.8	59.8	55.7
		MCDB	54.0	57.4	64.4	71.2	79.5	85.9	89.8	90.5	82.1	73.0	64.8	58.4
	5%	WB	47.7	49.9	54.0	59.0	67.2	72.7	75.2	75.3	68.9	63.6	57.3	52.2
		MCDB	50.2	54.7	60.9	68.1	77.4	83.0	86.0	87.2	79.7	70.3	61.8	54.6
	10%	WB	44.7	47.1	51.9	57.1	64.8	70.4	73.0	72.9	66.8	61.3	54.7	49.2
		MCDB	46.9	51.5	58.2	64.8	74.4	80.4	84.2	84.3	76.5	67.4	58.1	51.2
Mean Daily Temperature Range	5% DB	MDBR	12.3	15.8	18.1	20.4	23.5	24.3	25.3	25.5	22.6	18.3	15.2	12.1
		MCDBR	17.9	22.6	24.7	25.9	29.4	29.4	30.6	30.5	28.7	24.0	19.4	14.3
		MCWBR	13.3	14.9	14.8	14.7	15.5	13.7	13.8	13.5	13.5	12.7	11.7	10.8
	5% WB	MCDBR	16.3	20.3	22.3	23.8	26.6	27.9	28.6	28.5	26.2	21.6	17.5	13.2
		MCWBR	13.0	14.4	14.3	14.4	15.6	15.8	16.2	15.3	13.9	13.2	11.2	10.4
Clear-Sky Solar Irradiance	taub	0.346	0.378	0.419	0.469	0.443	0.444	0.447	0.455	0.433	0.400	0.373	0.351	
	taud	2.325	2.239	2.143	2.013	2.165	2.202	2.206	2.181	2.214	2.286	2.307	2.312	
	Ebn at Noon	247	257	259	254	265	265	262	256	252	246	234	233	
	Edh at Noon	28	35	43	53	47	45	44	44	40	33	28	26	
All-Sky Solar Radiation	RadAvg	489	737	1094	1511	1894	2057	2124	1910	1410	913	568	431	
	RadStd	64	82	95	158	136	158	131	89	109	93	63	62	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	+4.59	+5.51	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	+0.83	N/A	N/A	N/A	+1.22	+1.35	-162	-227	+164	N/A

Nomenclature: See separate page